MULTIMEDIA UNIVERSITY

STUDENT ID NO

MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER JULY/AUGUST 2025 (TERM 2520)

BAQ6323 – MANAGEMENT DECISION SCIENCE

(All sections / Groups)

8 OCTOBER 2025
9.00 a.m – 12.00 p.m
(3 Hours)

INSTRUCTIONS TO STUDENTS

1. This question paper consists of **FOUR (4) printed pages** excluding the cover page.
2. Answer **ALL** questions. All questions carry equal marks with the mark distribution for each question provided.
3. Only a non-programmable calculator is allowed to be used in this examination.
4. Please write all your answers in the provided Answer Booklet and ensure that all necessary workings are shown.

Question 1 (20 Marks)

(a) Tracksaws, Inc. makes tractors and lawn mowers. The firm makes a profit of RM30 on each tractor and RM30 on each lawn mower, and they sell all they can produce. The time requirements in the machine shop, fabrication and assembly are given in the following table:

Product	***Resource (hour)***		
	Machine shop	Fabrication	Assembly
Tractor	2	2	1
Lawn mower	1	3	0
Maximum availability	60	120	45

(i) Formulate the problem as a linear programming model. **(3 marks)**

(ii) Set up the initial simplex tableau by including the necessary slack variables. **(4 marks)**

The optimal tableau for the above problem is given below, where S_1, S_2 and S_3 are the slack variables for the machine shop, fabrication time and assembly time, respectively.

C_j	→	30	30	0	0	0	
↓	Solution Mix	x_1	x_2	S_1	S_2	S_3	Quantity
30	x_2	0	1	13/4	-0.15	0	30
30	x_1	1	0	-14/5	0.75	0	15
0	S_3	0	0	1.5	-28/3	1	15
	Z_j	30	30	13.5	18	0	1350
	$C_j - Z_j$	0	0	-13.5	-18	0	

(iii) How many tractors and lawn mowers should be produced to maximize profit and how much profit will they make? **(2 marks)**

(iv) How much could the hour in assembly be changed without changing the solution mix? **(1 mark)**

(b) Biggio's store has four employees available to assign to four departments in the store. The average daily hour for each employee in department is shown in the following table:

Employee	**Department (Hours)**			
	1	**2**	**3**	**4**
Annie	340	160	610	290
Beth	560	370	520	450
Carla	270	220	350	420
Debbie	360	190	630	150

Determine which employee to assign to each department and indicate the optimal assignment table. **(10 marks)**

Continued...

Question 2 (20 Marks)

Mancho Company manufactures electric fans at three factories to meet the demand of three distributors. The production capacity, the demand of the distributors and the transportation cost per unit in RM from each factory to each distributor are shown in the following tables:

	Distributors			
Factories	D	E	F	Supply
A	5	8	6	70
B	10	9	11	100
C	7	6	8	30
Demand	50	90	60	

Find an initial solution using Northwest Corner Method and determine the optimal distribution plan that minimizes the total distribution costs. **(20 marks)**

Question 3 (20 Marks)

Aisyah Electric is a company that provides all kinds of electrical works for commercial and housing area. Currently, the company has been awarded a new electrical wiring project. A list of activities and their shortest, most likely and longest time in days are given in the following table:

Activity	*Immediate Predecessors*	*Duration (Days)*		
		Shortest	*Most Likely*	*Longest*
A	-	4	6	8
B	-	2	4	6
C	-	1	2	3
D	C	6	7	8
E	B,D	2	4	6
F	A,E	6	10	14
G	A,E	1	2	3
H	F	3	6	9
I	G	10	11	12
J	C	14	16	18
K	H,I	6	8	10

(a) Determine the expected duration and variance for each activity. **(3 marks)**

(b) Represent the design involved in the form of an appropriate network of activities. **(2 marks)**

Continued…

(c) Determine the earliest start time, latest start time, earliest finish time, latest finish time and slack time. **(13 marks)**

(d) Identify the critical path of the new electrical wiring project. **(1 mark)**

(e) State the expected completion time of the new project. **(1 mark)**

Question 4 (20 Marks)

The ABC Co. is considering a new consumer product in view of intense competition. After a careful analysis of the market, the ABC Co. develops the following table:

Alternatives	States of nature		
	High demand (RM)	Fair demand (RM)	Poor demand (RM)
Add an assembly line	31 000	19 000	13 000
Plant addition	390 000	110 000	-30 000
Do nothing	0	0	0
Probabilities	0.6	0.3	0.1

(a) What decision would ABC Co. make if they applied Minimax Regret decision? **(4 marks)**

(b) Determine the best alternative if they applied Expected Opportunity Loss (EOL). **(5 marks)**

(c) Determine the best strategy using Expected Monetary Value (EMV). **(5 marks)**

(d) Select the best alternative for ABC Co. if they wanted to maximize the minimum payoff. **(3 marks)**

(e) What is the best decision using Maximax criterion? **(3 marks)**

Continued…

Question 5 (20 Marks)

A manager of an insurance company order letterhead stationery from an office products firm in boxes of 500 sheets. The company uses 6500 boxes per year. Annual carrying costs are RM 3 per box and ordering costs are RM25. The cost per box is RM 16 and the company is open 300 days in a year.

(a) Determine the optimal quantity of boxes to receive in each order. **(3 marks)**

(b) What is the annual holding cost? **(3 marks)**

(c) What is the annual ordering cost? **(3 marks)**

(d) Determine the total inventory cost associated with the optimal ordering policy including the purchase cost. **(2 marks)**

(e) If the lead time to receive an order is 3 days, how many boxes should be in hand when an order is placed? **(2 marks)**

(f) What is the optimal time between orders? **(2 marks)**

(g) The supplier firm will give a 5% discount if 500 boxes and above are ordered. Should the company consider the offer? **(5 marks)**

End of Questions.